Formula Sheet

Applied AI and Automation for Executives. The formulas you will meet in books, papers and vendor documents, with every symbol named.

How to use this sheet

Each row gives the standard name, the formula in standard notation, what each symbol means, a worked example with the numbers used in class, and the session that teaches it.
Different books use slightly different symbols. When you meet a formula elsewhere, look for the same pieces: the inputs, the weights, the loss and the step.
The worked numbers come from the course examples and recorded runs; the Session numbers refer to the slides and the Take-away Sheets.

Neurons and learning

Weighted sum of a neuron

$z = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b \quad = \quad \Sigma\, w_ix_i + b \quad = \quad w \cdot x + b$

Symbols: x: inputs. w: weights, how much each input counts. b: bias, shifts when the neuron switches on. z: the weighted sum.
Worked example: House: 2×100 m² + 10×3 bedrooms + 30 = 260 (UGX million).
Session 1

Activation functions

$\mathrm{ReLU}(z) = \max(0, z) \qquad \text{sigmoid } \sigma(z) = 1 / (1 + e^{\wedge}(-z))$

Symbols: f(z) is applied to the weighted sum. ReLU gives 0 for negative values and z otherwise. Sigmoid squeezes any number into 0 to 1.
Worked example: ReLU(−0.71) = 0; ReLU(4.58) = 4.58. σ(0) = 0.5; σ(4.58) = 0.99.
Session 1

A layer of neurons

$a = f(W\,x + b)$

Symbols: x: input vector. W: a matrix with one row of weights per neuron. b: one bias per neuron. a: the layer output (one number per neuron).
Worked example: Digit network: 64 inputs, 16 neurons: W is 16 × 64 = 1,024 weights, plus 16 biases.
Session 1

Loss (squared error and MSE)

$L = \tfrac{1}{2}\,(y - \hat{y})^2 \qquad MSE = (1/n)\,\Sigma\,(y_i - \hat{y}_i)^2$

Symbols: y: the true value. ŷ: the prediction (y-hat). n: number of examples. The ½ makes the derivative tidy; many texts omit it.
Worked example: Prices 220, 150, 310 against predictions 230, 150, 300: MSE = (100 + 0 + 100) / 3 = 66.7.
Session 0, 1

Cross-entropy loss (language models)

$L = -\ln p(\text{correct token})$

Symbols: p: the probability the model gave to the right next token. Smaller loss means a better guess.
Worked example: "days" got 18%: L = −ln 0.18 = 1.71. A perfect guess (p = 1) gives 0.
Session 5

Gradient descent

$w \leftarrow w - \eta \cdot \partial L/\partial w$

Symbols: η (eta): learning rate, the step size. ∂L/∂w: the gradient, how much the loss changes when w changes. ← means "is replaced by".
Worked example: x = 2, y = 3, w = 0.5, L = ½(y − wx)²: gradient = −x(y − wx) = −4. With η = 0.1: w = 0.5 + 0.4 = 0.9.
Session 0, 1

Backpropagation (chain rule)

$\partial L/\partial w = \partial L/\partial \hat{y} \cdot \partial \hat{y}/\partial z \cdot \partial z/\partial w$

Symbols: The chain rule: the effect of a weight on the loss is the product of the effects along the path (weight to sum, sum to prediction, prediction to loss), computed backwards from the output.

Worked example: For $\hat{y} = z = wx$ and $L = \frac{1}{2}(y - \hat{y})^2$: $(-(y - \hat{y})) \cdot 1 \cdot x = -2 \times 2 = -4$ at $w = 0.5$ ($y - \hat{y} = 2$, $x = 2$).

Session 0, 1

Examples and measuring

Gini impurity (decision trees)

$G = 1 - \Sigma\, p_k^2$

Symbols: p_k: the share of examples of class k in a group. 0 is a pure group.

Worked example: A group with 50% yes and 50% no: $G = 1 - (0.25 + 0.25) = 0.5$. A group of all yes: $G = 0$.

Session 2

Entropy and information gain

$H = -\Sigma\, p_k \log_2 p_k$ $\quad$ gain = H(parent) − weighted average H(children)

Symbols: The same idea as Gini: how mixed a group is. A split is chosen to maximise the gain.

Worked example: A 50/50 group: H = 1 bit. A 90/10 group: H = 0.47 bits.

Session 2

k-nearest neighbours and scaling

$\hat{y} = (1/k)\, \Sigma\, y_i$ (regression) $\quad$ majority vote (classification) $\quad$ $x' = (x - \min) / (\max - \min)$

Symbols: The k closest training examples decide. Scaling puts every feature on 0 to 1 so no feature dominates the distance.

Worked example: House prices of the 3 nearest sold houses 250, 270, 190: $\hat{y} = 710 / 3 = 237$. 80 m² on a 40 to 120 range: $(80 - 40) / 80 = 0.5$.

Session 2

Confusion-matrix measures

accuracy = (TP + TN) / (TP + TN + FP + FN) $\quad$ precision = TP / (TP + FP) $\quad$ recall = TP / (TP + FN)

Symbols: TP: true positives, FP: false positives, FN: false negatives, TN: true negatives. Recall is also called sensitivity or true positive rate.

Worked example: Smoke alarm TP 16, FP 34, FN 4, TN 946: accuracy 962/1000 = 96.2%; precision 16/50 = 32%; recall 16/20 = 80%.

Session 0, 3

F1, false positive rate, ROC and AUC

$F1 = 2PR / (P + R) = 2 / (1/P + 1/R)$ $\quad$ FPR = FP / (FP + TN)

Symbols: F1 is the harmonic mean of precision (P) and recall (R). The ROC curve plots TPR (recall) against FPR for every threshold; AUC is the area under that curve (1 is perfect, 0.5 is a coin flip).

Worked example: P = 0.32, R = 0.80: F1 = 0.46. FPR = 34 / 980 = 3.5%.

Session 0, 3

k-fold cross-validation

$\text{score} = (1/k)\, \Sigma\, \text{score}_i$

Symbols: Split the data into k parts; train on k − 1 parts, test on the held-out part; rotate; average the k scores.

Worked example: Session 2 claims: fold scores 44/48, 44/48, 45/48, 45/48, 45/48 (0.917, 0.917, 0.938, 0.938, 0.938) give 4.646 / 5 = 0.929.

Session 2

Words and language models

TF-IDF

$\text{tf-idf}(t, d) = \text{tf}(t, d) \times \log(N / \text{df}(t))$ $\quad$ scikit-learn: $\text{idf} = \ln[(1 + N) / (1 + \text{df})] + 1$

Symbols: t: a term. d: a document. tf: how often t appears in d. N: number of documents. df: how many documents contain t.

Worked example: N = 40, a term in 4 messages: idf (textbook) = log(40/4) = log 10; a term in 22 messages: log(40/22) = 0.6. Rare terms weigh more.

Session 4

Distances and similarity

$$\text{Euclidean } d = \sqrt{\Sigma(a_i - b_i)^2} \quad \text{Manhattan } d = \Sigma|a_i - b_i| \quad \text{dot } a \cdot b = \Sigma\, a_i b_i \quad \cos\theta = a \cdot b \,/\, (\|a\|\,\|b\|)$$

Symbols: a, b: vectors (lists of numbers). $\|a\| = \sqrt{\Sigma\, a_i^2}$ is the length. Cosine ignores length and compares direction; cosine distance = 1 − cos θ. Also called L2 and L1 distance and inner product.

Worked example: a = (3, 4), b = (6, 8): Euclidean = 5; Manhattan = 7; dot = 50; cos θ = 50 / (5 × 10) = 1.00.

Session 4, 9

Softmax and temperature

$$p_i = e^{(z_i/T)} \,/\, \Sigma_j\, e^{(z_j/T)}$$

Symbols: z: the logits (raw scores) for every token. T: temperature (T = 1 is plain softmax; lower T sharpens, higher T flattens). The p sum to 1.

Worked example: Logits 3, 2, 0: p = 70.5%, 25.9%, 3.5%. At T = 2: 55%, 33%, 12%.

Session 5

Language model and embeddings

$$P(x_1 \ldots x_n) = \Pi\, P(x_t \mid x_1 \ldots x_{t-1}) \qquad x_t = E[\text{token}_t] + \text{position}_t$$

Symbols: The model gives each next token a probability given the earlier ones; the sequence probability is the product. E: the embedding matrix (one row of numbers per token). Position information is added.

Worked example: GPT-2 small: E has 50,257 rows of 768 numbers (38,597,376 numbers).

Session 5

Scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}(\, Q\, K^{\mathsf{T}} / \sqrt{d_k}\,)\, V$$

Symbols: Q (queries), K (keys), V (values): matrices made from the token vectors. d_k: the key length. Each token scores every other token (QK^{T}), softmax turns scores into weights, and the output is a weighted blend of the values.

Worked example: See the Session 5 toy: scores turn into shares 9%, 67%, 24%; the output blends the values with those shares.

Session 5

Perplexity

$$\text{perplexity} = \exp(\text{ average cross-entropy per token })$$

Symbols: A standard way to report how well a language model predicts text; lower is better.

Worked example: An average loss of 1.71 gives perplexity 5.5.

Session 6

Size and adaptation

Model memory

$$\text{memory} \approx \text{parameters} \times \text{bytes per parameter}$$

Symbols: FP32 = 4 bytes, FP16 = 2, INT8 = 1, 4-bit = 0.5. Weights only; running a model needs more memory for working data.

Worked example: GPT-2 small: 124,439,808 × 4 bytes ≈ 498 MB in FP32; 7 billion parameters at FP16 = 14 GB, at 4-bit = 3.5 GB.

Session 7

Quantisation

$$q = \text{round}(x / s) + z \qquad \hat{x} = s \cdot (q - z)$$

Symbols: x: the original number. s: scale. z: zero point. q: the stored whole number. x̂: the value recovered (close to x, not identical).

Worked example: x = 0.37, s = 0.01, z = 0: q = 37, x̂ = 0.37. x = 0.374: q = 37, x̂ = 0.37 (a small rounding error).

Session 7

LoRA (low-rank adaptation)

$$W' = W + B\, A \qquad \text{trainable numbers} = r\,(d + k) \text{ instead of } d \times k$$

Symbols: W: a frozen d × k weight matrix. B: d × r. A: r × k. r: the rank (small). Only B and A are trained. The real method scales BA by a constant.

Worked example: d = 768, k = 2,304, r = 8: r(d + k) = 24,576 against d × k = 1,769,472.

Session 9

Retrieval and the context budget

retrieve the k chunks with the largest cos θ prompt + retrieved + answer ≤ context window

Symbols: Each chunk and the question are embedded; the k most similar chunks go into the prompt. The total tokens must fit the model's context window.

Worked example: Question 94 + 3 chunks 104 + answer limit 300 = 498 tokens, against 94 + 1,074 + 300 = 1,468 for the whole handbook.

Session 9

Evaluation accuracy and expected cost

accuracy = correct / total expected cost = P(error) × cost of an error

Symbols: Score a model or a prompt on a fixed evaluation set; weigh how often it errs by how much an error costs.

Worked example: Baseline answering "not stated" scores 2 / 10 = 20%.

Session 8, 10

Agents, governance and plan

Risk score and approval time

risk = likelihood × impact approval time T = n × t

Symbols: Likelihood and impact are each rated (for example 1 to 5). n: number of approvals, t: minutes per approval. Least privilege: granted permissions ⊆ needed permissions.

Worked example: Flat tyre: 4 × 2 = 8. Data leakage: 3 × 5 = 15. Three approvals at 2 minutes: 3 × 2 = 6 minutes.

Session 11, 13

Hours saved, net benefit, payback and ROI

$H = n \times (t_0 - t_1) / 60$ net benefit = benefits − costs payback = set-up cost / net benefit per month ROI = net benefit / cost × 100%

Symbols: n: runs a month. t_0, t_1: minutes before and after. Payback is in months.

Worked example: n = 4, t_0 = 90, t_1 = 10: H = 5.3 hours. Kettle: saving 20,000 a month, cost 60,000: payback 3 months; first-year net 180,000; first-year costs 180,000; ROI 100%.

Session 12

Total cost of ownership

TCO = set-up + months × monthly fee

Symbols: All the money over the whole contract, not only the first month.

Worked example: Set-up 0, 36 months at USD 2,400: 36 × 2,400 = USD 86,400.

Session 13

Data quality and readiness

completeness = filled / all × 100% uniqueness = distinct / all readiness = Σ points (Green 2, Amber 1, Red 0) change = (new − old) / old

Symbols: Completeness counts non-missing values; uniqueness counts distinct records; readiness sums six area ratings out of 12.

Worked example: NVF: (60 − 2) / 60 = 96.7% complete; 59 / 60 = 0.983 unique; 2+1+1+1+0+1 = 6 of 12; score 6 to 9: +50%.

Session 14